MIL-R-23139B (WEP)
18 August 1965
Supersedes
MIL-R-23139A (WEP)
7 May 1963

CODE IDENT
10001

MILITARY SPECIFICATION

ROCKET MOTORS, SURFACE LAUNCHED, DEVELOPMENT AND QUALIFICATION REQUIREMENTS FOR

This specification has been approved by
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification defines the design requirements and acceptance test for surface launched rocket motors.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-7179	Finishes and Coatings, General Specifications for
MIL-R-22713 (WEP)	Rocket Motors, Forty Foot Drop Test
MIL-D-23660 (WEP)	Data, Technical, for Rocket Motors
MIL-P-24014 (WEP)	Preclusion of Hazards from Electromagnetic Radiation to Ordnance, General Requirements for

FSC 1336

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-447	Definitions of Interchangeable Substitute and Replacement Items
MIL-STD-709	Ammunition Color Coding
MS 33586	Metals, Definitions of Dissimilar

PUBLICATIONS

Military

MIL-HDBK-5	Metallic Materials and Elements for Flight Vehicular Structures

Department of the Navy
Bureau of Naval Weapons

Instruction 8020.3	Explosive Hazards, Classification Procedures

(When requesting any of the above documents, give the title and complete designation of the item as shown above. Copies of specifications, standards and publications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS) 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents may be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications. -The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply to the extent specified herein.

Department of Commerce
National Bureau of Standards

H28	Screw-thread Standards for Federal Services

(Request for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

Interstate Commerce Commission

49 CFR 71-90	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition, current revision) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71-90, current revision".)

3. REQUIREMENTS

3.1 General requirements. -Rocket motors covered by this specification shall be designed to ensure maximum safety, reliability, efficiency, ease of operation and handling and economical production. The rocket motor developer shall be responsible for coordinating with the missile systems designer to assure that the rocket motor is compatible with the interface dimensions, launcher and handling equipment and the missile performance.

3.2 General design requirements. -The following features shall be included in, but not limited to, the design of the rocket motor.

3.2.1 Interchangeability. -All component parts of the rocket motor shall be designed to be interchangeable as specified in MIL-STD-447 except when specifically otherwise authorized by the Bureau of Naval Weapons, RMMP-3.

3.2.2 Replaceability. -All major rocket motor components which will be assembled in the Navy Depots shall be replaceable.

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3.2.3 Threads. -All threads shall conform to NBS Handbook H28.

3.2.4 Materials. -All materials used in propellant formulation, hardware components, adhesives, etc. shall be of existing government specifications where possible. For materials not covered by existing government specifications see requirements of MIL-D-23660 (WEP).

3.2.5 Dissimilar metals. -Contacts between dissimilar metals as defined in Military Standard MS 33586 which cause a significant corrosion rate shall be avoided. Where such contact is unavoidable, the parts shall be insulated as recommended in MIL-F-7179.

3.2.6 Weather seal. -Rocket motor design shall include an airtight weather seal at the exit nozzle area to prevent water or moisture from entering the rocket motor. The weather seal shall be effective after all methods of transportation including a pressure differential that may occur during air transportation.

3.2.7 Wiring. -Wiring and terminal design shall include protection against moisture, fungus, spray and mechanical hazards to prevent corrosion and shorting of circuits. It shall be possible to connect and disconnect the electrical connectors and to flex the conductors without damage within the storage temperature range specified for the motor.

3.2.8 Pressure testing. -The rocket motor design shall include means to pressure test the loaded rocket motor, including the igniter, at 7 to 10 psig dry air to verify the effectiveness of the O-ring seals and weather seals.

3.2.9 Fillet radii. -Small fillet radii shall be avoided in the rocket motor to avoid stress concentration.

3.2.10 Castings. -Castings shall not be used on parts which are stressed 50 percent on more of the material yield strength. Drawings shall state clearly when castings are not allowed.

3.2.11 Safety. -The design shall include means to make the rocket motor nonpropulsive during transportation, storage and handling such that the effective thrust is less than the weight of the loaded motor unless otherwise authorized by the Bureau of Naval Weapons, RMMP-3.

3.2.12 Igniter.

3.2.12.1 Dual wiring. -The igniter design shall include two (2) parallel firing circuits including two (2) initiators for increased reliability. Either circuit shall be capable of firing the igniter.

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3.2.12.2 Ignition current. -The firing current of the igniter shall be compatible with the electrical characteristics available from the applicable launcher. The minimum no-fire current through the igniter firing circuit shall be one (1) ampere for 30 seconds minimum period.

3.2.12.3 Rocket motors which are to be ignited on the launcher shall be provided with a remote-controlled, reversible igniter arming system. Rocket motors which are to be ignited in flight shall be provided with an arming system which shall arm the ignition system after the missile is launched and after it has reached a predetermined safe arming distance from the launcher.

3.2.12.4 Manual arming and disarming of the igniters of rocket motors which are to be ignited on the launcher shall be possible when the rocket motor and missile are mated or assembled as a complete round.

3.2.12.5 The rocket motor igniter shall incorporate a mechanically operated safety mechanism which shall provide for the following conditions and shall include provisions for visual determination of the selected condition of the ignition system in an assembled round:

SAFE: The initiator(s) shall be mechanically isolated from the igniter charge and electrically isolated from the igniter wiring.

ARMED: The igniter explosive train shall be aligned prior to, or at the same time as, completion of the electrical circuit to the initiators. The electric path between the firing contacts and initiator(s) shall be continuous.

TEST: The initiator(s) shall be mechanically isolated from the igniter charge, and the electrical path between the firing contacts and the initiator(s) shall be continuous.

3.2.12.6 The igniter safety mechanism shall be capable of being returned to the SAFE condition from the ARMED condition.

3.2.12.7 A mechanical resistance shall be included in the igniter safety mechanism to prevent accidental movement of the mechanism from one position to another position, except where the launcher arming device permits missile launch while holding the igniter mechanism in the armed position.

3.2.12.8 Failure of the igniter safety mechanism to lock in the ARMED position while undergoing operational movement shall result in the automatic return of the mechanism to the SAFE position.

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3.2.12.9 Electromagnetic radiation hazard. -The igniter shall be designed to meet the preclusion of Hazards from Electromagnetic Radiation to Ordnance (HERO) in accordance with MIL-P-24014 (WEP).

3.2.13 Propulsion system. -Solid propellant and liquid fuel formulations which are different from formulations previously accepted for service use shall be approved by the Bureau of Naval Weapons, RMMP-3. For all new propellant fuel formulations, the following characteristics, as a minimum, shall be determined, and complete internal documentation on such determinations shall be made available to the Bureau of Naval Weapons, RMMP-3, prior to approval of the new formulation.

3.2.13.1 Sensitivity to initiation by shock, vibration, impact or heat.

3.2.13.2 Effects of aging and environmental extremes on sensitivity.

3.2.13.3 Type and amount of toxic materials in original formulations and in reaction products.

3.2.13.4 Detonability.

3.2.13.5 Thermal stability.

3.3 Program phases. -The design acceptance shall be conducted by the following phase:

I) Development

A) Design stage
B) Pre-Flight Rating Tests (PFRT) Stage

II) Qualification

A) Performance Stage
B) Hazard Classification Stage

Prior to commencing work on Phase I, an internal master development plan shall be prepared by the contractor and approved by the Bureau of Naval Weapons, RMMP-3, at which time it shall become a part of this specification. The minimum outline for the master development plan is presented in Section 6.2. Prior to commencing work on Phase II, an internal Model Specification shall be prepared by the contractor and approved by the Bureau of Naval Weapons, RMMP-3, at which time it shall become a part of this specification. Tests to be included in the Model Specification are presented in Section 6.3.

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3.3.1 Development phase. -The development phase shall be designed to provide the information necessary to complete the documentation required for Phase I under MIL-D-23660 (WEP).

3.3.1.1 Design stage. -The structural integrity of the rocket motor shall be verified by applicable load tests to demonstrate that the proposed rocket motor design meets the requirements of the missile system.

3.3.1.2 Pre-Flight Rating Test (PFRT) stage. -The rocket motor shall successfully pass the tests as outlined in the master development work plan to demonstrate that the rocket motor can be safety handled through the qualification tests and the experimental missile flights.

3.3.2 Qualification phase. -The Model Specification shall be designed to provide the information necessary to complete the documentation required for Phase II under MIL-D-23660 (WEP).

3.3.2.1 Performance stage. -The performance of the rocket motor shall be compatible with the missile system requirements.

3.3.2.2 Hazard classification stage. -The hazard classification shall demonstrate that the rocket motor is not a safety hazard under normal operating and handling environments.

3.4 Vibration. -The rocket motor shall meet the specified performance and structural requirements specified in the Model Specification after being subjected to the vibrations induced by transportation, shipboard stowage and the launcher.

3.4.1 Vibration requirements. -The rocket motor shall withstand vibration at resonance over the frequency range of 5-300 cycles per second. If there is more than one resonance over any frequency range the specified time may be evenly divided among the resonances. If no resonances are observed, the rocket motor shall be vibrated at 40, 100 and 300 cps for the specified time. The vibration time shall be as follows:

cps	g (min)	hrs (min)
5-40	3.5	72
41-100	1.5	72
101-300	0.5	12

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3.4.2 Shipboard stowage and launcher. -The rocket motor shall withstand vibration of Type 1 as specified in MIL-STD-167. The vibration temperatures shall be 0°F and 120°F.

3.5 Auto ignition. -The propellant and igniter shall not igniter while the motor is exposed to 350°F temperature for two (2) hours minimum.

3.6 Exhaust. -The rocket motor shall not eject particles or residue harmful to launcher operation.

3.7 Smoke. -The propellant shall be designed so that the amount of smoke is low.

3.8 Operating temperature. -The minimum firing temperature range shall be from +20°F to +110°F. For qualification of the rocket motor, the firing limits shall be from +10°F to +120°F.

3.9 Storage temperature. -The minimum storage temperature range shall be from 0°F to +120°F.

3.10 Age limit. -The propellant grain design shall provide satisfactory operation for a minimum period of five (5) years, under the service environment and conditions. To estimate the minimum grain life, full scale rocket motors shall be subjected to continuous storage at the storage temperature extremes (3.9) for periods up to six months. The rocket motors shall be withdrawn from storage and fired as specified in the Model Specification.

3.11 Factor of safety. -The rocket motor shall be designed to withstand the Maximum Expected Operating Pressure (MEOP) of the propellant. The minimum factor of safety for the pressure chamber shall be 1.15 X MEOP. For other structural members of the rocket motors, the minimum factor of safety shall be 1.20 times the loads specified in the Design Approval Drawings. Stress calculations for the factor of safety shall be based on the maximum design loads, the yield strength of material and the minimum material conditions in accordance with MIL-HDBK-5.

3.12 Forty foot drop test. -The rocket motor shall meet the requirements of the forty foot drop test specified in MIL-R-22713 (WEP).

3.13 Shipping and identification. -All units shipped outside the contractor's facility shall be identified in accordance with the requirements of MIL-STD-709.

3.14 Data requirements. -No data is required by this specification or by applicable documents referenced in Section 2, unless specified in the contract or order.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. -The contractor is responsible for all inspection and quality control needed in the design and production of rocket motors submitted for qualification testing. Acceptance test information, as well as information on any rocket motor or components rejected during acceptance, shall be submitted as part of the qualification test information. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplier and services conform to the prescribed requirements.

4.2 Classification of tests. -Inspection and testing of rocket motors shall be classified as follows:

4.2.1 Development test. -The development tests shall demonstrate that the rocket motor design and performance is suitable for flight testing.

4.2.2 Qualification test. -The qualification tests shall demonstrate that the rocket motor design and performance is suitable for service use.

4.3 Tests.

4.3.1 Development tests. -Development tests shall be conducted in accordance with the approved master development plan as required in 3.3.

4.3.2 Qualification tests. -Qualification tests shall be conducted in accordance with the approved Model Specification as required in 3.3.

4.4 Test firings. -Unless otherwise specified by the Bureau of Naval Weapons, RMMP-3, all test firings shall be conducted at the following temperatures:

Low	$+10^{\circ}F \pm 5^{\circ}F$
Ambient	$+77^{\circ}F \pm 5^{\circ}F$
High	$+120^{\circ}F \pm 5^{\circ}F$

4.5 Acceptance criteria. -Acceptance shall be based on all motors tested meeting all the requirements of the qualification tests.

5. PREPARATION FOR DELIVERY

5.1 Rocket motors. -All rocket motors shipped for testing outside of contractors facility shall be in non-propulsive condition, unless otherwise authorized by the Bureau of Naval Weapons, RMMP-3.

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5.1.1 Preservation and packaging.

5.1.1.1 Level A.

5.1.1.1.1 Cleaning, drying, preservation. -Units shipped for testing outside the contractors facility shall be cleaned, dried, and preserved as required on applicable drawings or as specified by the procuring activity.

5.1.1.1.2 Unit packaging. -Igniters and other components of rocket motors which are shipped separately outside of the contractor's facility shall be packaged in accordance with applicable drawings or as specified by the procuring activity.

5.1.1.1.3 Intermediate package. -Not applicable.

5.1.1.2 Level B. -Not applicable.

5.1.1.3 Level C. -Units shipped for testing outside of the contractor's facility shall be packaged in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.1.2 Packing.

5.1.2.1 Level A.

5.1.2.1.1 Exterior containers. -Units shipped outside of the contractor's facility shall be packed in accordance with applicable crate or container drawings.

5.1.2.2 Level B. -Not applicable.

5.1.2.3 Level C. -Units shipped for testing outside of the contractor's facility shall be packed in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.2 Marking.

5.2.1 Special marking. -Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.2.2 Normal marking. -In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.2.3 Unit marking. -Each rocket motor shipped outside of the contractor's facility shall be clearly marked with the following information:

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(1) Model designation (rocket motor and igniter)

(2) Operating temperature range

(3) Storage temperature range

(4) Manufacturer's name

(5) Serial number

(6) Handling instructions and warning as applicable

(7) Inspection date

(8) Loading date

(9) Loaded center of gravity

6. NOTES

6.1 Intended use. -This specification is intended for use in the development of a rocket motor for surface launch applications.

6.2 Development phase. -The following is the minimum testing and analysis that is to be incorporated in the master development plan required in 3.3 for the development phase (Phase I).

6.2.1 Structural integrity. -Structural integrity of the rocket motor shall be verified by applicable load tests to demonstrate that the proposed rocket motor design meets the requirements of the missile system.

6.2.2 Test firings. -The quantity of units fired is advisory. Small scale motors and twenty (20) full scale rocket motors shall be manufactured for static test firings. If the initial full scale testing is done by using heavy weight motor parts, at least ten (10) rocket motors shall be made using standard flight weight parts. The motors shall be static test fired at low, ambient (77°F), and high qualification temperatures (see 4.4).

6.2.3 Analysis and testing during the design stage shall be designed to derive information that will include, but not be limited to the following:

6.2.3.1 Performance characteristics including:

-pressure-time curves at low, ambient, and high operating temperatures
-thrust-time curves at low, ambient, and high operating temperatures

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- total impulse at low, ambient, and high operating temperatures
- radar attenuation characteristics

6.2.3.2 Propellant composition and properties including:

- formulation
- specific impulse (theoretical and delivered)
- burning rate at ambient temperature
- temperature coefficient of pressure
- pressure exponent
- flame temperature
- strain at maximum stress
- modulus of elasticity

6.2.3.3 Igniter design and firing circuit including:

- igniter charge formulation
- bridge wire resistance
- maximum no fire current
- all fire current

6.2.3.4 Rocket motor data including:

- weight, including breakdown on major components
- center of gravity (loaded and burnt)
- list of proposed principle materials
- wall thickness of combustion chamber
- mass fraction
- volumetric efficiency
- port-to-throat ratio
- nozzle expansion ratio
- throat area
- assembly dimensions
- correlation dimensions as required

6.2.4 Pre-Flight Rating Tests (PFRT).

6.2.4.1 Types of tests and number of motors. -The types of tests and the number of motors to be tested in the Pre-Flight Rating Tests (PFRT) shall be as indicated in 6.2.4.1.1 and 6.2.4.1.2. Any departure from the PFRT test plan must have prior approval of the Bureau of Naval Weapons, RMMP-3.

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6.2.4.1.1 Test firings.

(1) Six (6) motors (two at each temperature) shall be static fired at low (10°F); ambient (77°F) and high (120°F).

(2) Three (3) motors shall be subjected to three (3) full temperature cycles between -10°F and 130°F. Rocket motors shall be inspected for visual damage after cycling. The three (3) motors shall be static fired - one at low (10°F), one at ambient (77°F) and one at high (120°F). At the completion of static test firing, the motors shall be visually inspected for damage.

(3) One (1) motor shall be fired at ambient temperature (77°F) in a centrifuge in the longitudinal position simulating the launch and flight loads.

6.2.4.1.2 Flight test.

(1) Four (4) rocket motors for use in Rocket Test Vehicles (RTV's) shall be forwarded to an activity designated by the Bureau of Naval Weapons, RMMP-3, for experimental missile flight tests.

6.3 Qualification. -The following are the minimum tests that are to be incorporated into the Model Specification required in 3.3 for the qualification phase (Phase II).

6.3.1 Rocket motors submitted for qualification testing shall be produced to the same design as the rocket motors that successfully passed the development phase tests.

6.3.2 Performance stage. -Type of tests and number of motors to be tested in the performance stage shall be as indicated in 6.3.2.1, 6.3.2.2 and 6.3.2.3. Any departure from the qualification test plan must have prior approval of the Bureau of Naval Weapons, RMMP-3.

6.3.2.1 Test firings. -Fifty-four (54) rocket motors shall be test fired as follows:

6.3.2.1.1 Centrifuge firings. -Six (6) motors shall be fired in a centrifuge under the lateral acceleration load. Of the six motors, two (2) shall be fired at each qualification temperature (low, ambient, high) of the rocket motor (see 4.4).

6.3.2.1.2 Temperature gradient firings. -Six (6) temperature gradient motors shall be fired at the maximum propellant temperature gradient. Of the six motors, three (3) shall be conditioned to 20°F and then immediately placed in a 110°F conditioning oven for a period which will cause a maximum temperature gradient to exist within the propellant grain. The remaining three (3) motors shall be similarly conditioned from 110°F to 20°F.

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6.3.2.1.3 Accelerated aging firings. -Eight (8) motors shall be static test fired after being stored for the periods and at the temperatures given in the table below. Motors shall be visually inspected during storage and prior to firing, for visual damage.

Number of Units	Storage Temperature	Storage Period	Static Firing Conditions (see 4.4)
3	120°F	3 mos.	2 fired at high qualification temp. 1 fired at low qualification temp.
3	120°F	6 mos.	2 fired at high qualification temp. 1 fired at low qualification temp.
2	0°F	6 mos.	1 fired at high qualification temp. 1 fired at low qualification temp.

6.3.2.1.4 Sequential test firings. -Seventeen (17) motors shall be static test fired at the temperatures and sequences shown in Table 1.

6.3.2.1.4.1 Temperature cycling. -Motors requiring temperature cycling in accordance with Table 1 shall be cycled to simulate the most abrupt temperature changes expected in rocket motor handling. The motors shall be conditioned at the specified temperature for a minimum of 24 hours. The motors shall be visually inspected for damage after cycling and prior to statically firing. The motors shall be temperature cycled as follows:

(a) -10°F, 130°F, -10°F, 130°F, -10°F, 130°F, Firing Temperature

6.3.2.1.4.2 Vibration test. -Motors requiring vibration in accordance with Table 1 shall be vibrated in accordance with 3.4 at both in the low and high temperatures. The motor shall be visually inspected for damage after testing.

6.3.2.1.4.3 Drop test. -Motors requiring drop test in accordance with Table 1 shall be dropped on concrete as follows:

	*Crated Motors	
Drop Distance (feet)	Temp (°F)	Process
3	0	One end lifted and dropped
1	0	Dropped on side
3	+70	Dropped on side

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*Uncrated Motors

Drop Distance (feet)	Temp (°F)	Process
6	0	One end lifted and dropped
3	0	Dropped on side
6	+70	Dropped on side

* Only one of the two tests shall be used depending upon the nature of the external attachments on the motor.

TABLE I
SEQUENTIAL TEST FIRINGS

Motor No.	*Test Sequence	Firing Temperature, °F
1	A	+ 20
2	A	+130
3	B	+ 20
4	B	+ 80
5	B	+130
6	C	+ 20
7	C	+ 80
8	C	+130
9	BA	+ 20
10	BA	+130
11	CB	+ 20
12	BC	+ 80
13	CB	+130
14	CA	+ 20
15	AC	+130
16	CBA	+ 20
17	ABC	+130

*where: A = Temperature cycling in accordance with 6.3.2.1.4.1
B = Vibration in accordance with 6.3.2.1.4.2
C = Drop test in accordance with 6.3.2.1.4.3

6.3.2.1.5 Environmental testing. -Two (2) motors shall be static fired (one each of the high and low qualification temperatures (4.4) after being submitted to environmental testing. The following tests shall be included:

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(1) Rain

(2) Sand and dust

(3) Salt spray

6.3.2.1.6 Reliability testing. -Fifteen (15) motors shall be static test fired, five (5) each at low, ambient and high motor qualification temperatures (see 4.4) to test the motor reliability.

6.3.2.2 Flight testing. -Unless otherwise specified, twenty-five (25) rocket motors shall be flight tested as specified by the Bureau of Naval Weapons, RMMP-3.

6.3.2.3 Additional tests. -In addition to verification of performance requirements, the following characteristics shall be determined in the static firings or flight tests as applicable:

- exhaust blast pattern
- thrust alignment
- propellant fragment ejection
- smoke
- exhaust residue inspection
- motor skin temperatures during and after firings
- thrust neutralizer firings
- roll unbalance measurements
- radar signal attenuation

6.3.2.4 Missile coordination. -Unless otherwise specified, five (5) inert motors shall be forwarded to the missile contractor for coordination purposes. The motors shall be empty or inert loaded.

6.3.2.5 Structural tests. -The structural integrity of rocket motors shall be verified by applicable load tests.

6.3.3 Hazard classification stage. -Hazard classification shall be conducted by the contractor in accordance with BuWeps Inst 8020.3.

6.4 Definitions.

6.4.1 Design approval drawings. -The design approval drawings provide sufficient initial information to enable the design activities to proceed with the design and development. The design approval drawings are superseded by production drawings at the beginning of rocket motor production.

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6.4.2 Model Specification. -Model Specification is a document produced by the rocket motor contractor and contains all the rocket motor requirements and the detailed qualification program to demonstrate that the rocket motor meets the requirement.

6.5 Shipping container design. -This specification does not include the design, development, test and documentation of containers and handling equipment for service shipment and handling. It covers that packaging, packing and handling equipment required only for shipment and handling of the rocket motors used in the development and tests contained herein.

6.6 Facilities. -Many of the tests required to evaluate the rocket motor environment, performance and hazard classification involve specialized facilities. The Government maintains facilities for these purposes and are available under contract. A list of facilities are available through the Bureau of Naval Weapons, Code RMMP-3.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-23139B(WP) ROCKET MOTORS, SURFACE LAUNCHED DEVELOPMENT & QUALIFICATION REQ FOR.

ORGANIZATION *(of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Washington, D. C. 20360

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BUREAU OF NAVAL WEAPONS (RREN-5)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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